Pattern Making Studies

A Parametric Approach to Flat Pattern Drafting with R

Your Name

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Contents

Preface

About This Book

This book documents my journey learning pattern making through a parametric, code-based approach. Rather than traditional paper drafting, I translate pattern making instructions into R code, creating:

- Reproducible patterns that adjust to any measurement set
- Fully annotated diagrams with all construction points visible
- A searchable reference of drafting techniques
- A record of errors, corrections, and learning insights

Why Parametric Pattern Making?

Traditional pattern making relies on manual drafting for each size and design. This approach:

1. **Eliminates manual grading**: Change measurements, regenerate the pattern
2. **Preserves design logic**: Every calculation is documented in code
3. **Enables experimentation**: Test design changes instantly
4. **Creates a knowledge base**: Build upon previous work systematically

Tools Used

- **R** and **R Markdown/Bookdown**: Documentation and computation
- **SeamlyMe**: Body measurement management
- **FreeSewing Notation**: Standard pattern marking symbols
- **Git/GitHub**: Version control and sharing

How to Use This Book

The book follows a structured learning path:

1. **Chapter 0**: Notation standards and conventions
2. **Chapter 1**: Measurement systems and file formats
3. **Chapters 2a-2f**: Basic blocks for different garments
4. **Chapter 3**: Design variations from basic blocks
5. **Chapter 4**: Real projects with iterations and corrections
6. **Chapter 5**: Reference tables and formulas

Prerequisites

- R (>= 4.0) and RStudio
- LaTeX distribution (for PDF output)
- Basic understanding of garment construction
- Optional: SeamlyMe for measurement management

Acknowledgments

- FreeSewing.org for the notation system
- Traditional pattern making textbooks
- Open source community

1. Notation Standards and Conventions

This chapter defines all notation, terminology, and standards used throughout this pattern making documentation. Following these conventions ensures consistency across all pattern blocks and variations.

1.1. Measurement System

All measurements in this book use the **metric system** exclusively.

Unit	Abbreviation	Precision	Usage
Centimeter	cm	0.1	Body measurements, pattern dimensions, ease
Millimeter	mm	0.5	Notches, seam allowances, small details
Meter	m	0.01	Fabric length estimation only

Critical rule: Never mix measurement systems. All formulas assume centimeter inputs and return centimeter outputs.

1.2. Coordinate System

All patterns are drafted on a 2D Cartesian coordinate system following textile industry conventions:

- **Origin (0,0)**: Bottom-left corner of the foundation rectangle (point D or C depending on orientation)
- **X-axis**: Horizontal, positive values extend to the right (pattern width)
- **Y-axis**: Vertical, positive values extend upward (pattern length/height)

```
Y-axis (Height)
^
|
|   A(0, height) ----------- B(width, height)
|   |                        |
|   |    FRONT      BACK     |
|   |                        |
|   D(0, 0) --------------- C(width, 0)
|
+-------------------------------------------> X-axis (Width)
Origin (0,0)
```

1.3. Point Naming Conventions

Construction points follow traditional pattern drafting conventions with letter designations:

1.3.1. Primary Structure Points (Foundation Rectangle)

Point	Description	Typical Position
A	Top-left corner	(0, height)
B	Top-right corner	(width, height)
C	Bottom-right corner	(width, 0)
D	Bottom-left corner	(0, 0)

1.3.2. Construction Division Points

Point	Description	Derivation
E, F	Side seam offset	A + offset, D + offset
G, H	Center/Armscye division	Midpoint of EB, FC
I, J, K, L	Front piece divisions	Divide AG into 4 equal parts

1.3.3. Pattern Feature Points

Category	Points	Represents
Neckline	N, Decote points	Front and back neckline curves
Shoulder	M, P, I, L	Shoulder seam endpoints
Armscye	Q, R, S, T, U, V, X, Y, Z	Sleeve cap curve control points
Darts	Various	Dart apex and legs
Waist/Hip	W, Hip points	Waistline and hip curve points

1.4. Line Types and Meanings

Based on the FreeSewing Notation System, adapted for R/ggplot2:

1.4.1. Structural Lines

Line Type	R Linetype	Color	Purpose
Seam line	`solid`	Black (#212121)	Actual sewing line
Seam allowance	`dashed`	Grey (#757575)	Fabric cutting line
Construction line	`dotted`	Light grey	Temporary guide lines
Grainline	`dotted` + arrow	Green (#388E3C)	Fabric grain direction
Cut-on-fold	`twodash` + arrows	Red (#D32F2F)	Place on fabric fold
Center line	`dashed`	Grey	Center front/back

1.4.2. Annotation Lines

Line Type	R Linetype	Color	Purpose
Dimension	`dotted` thin	Blue (#1565C0)	Measurement callouts
Note	`dashed` thin	Dark grey	Construction notes
Mark	`solid` thin	Various	Placement indicators
Contrast	`solid` colored	Cyan (#00BCD4)	Highlight specific areas
Help	`dotted` thin	Light grey	Assembly guides

1.5. Notches and Markings

1.5.1. Notch Types (FreeSewing Standard)

- **Front notch**: Circle with dot center – indicates front of garment
- **Back notch**: Circle with cross – indicates back of garment

1.5.2. Construction Marks

Symbol	R Implementation	Meaning
(.)	`add_notch(type="front")`	Front orientation
(+)	`add_notch(type="back")`	Back orientation
(o)	`add_button()`	Button placement
(	)	`add_buttonhole()`
−x−	`add_bartack()`	Reinforcement stitch
^	`add_grainline()`	Grain direction (arrow up)

1.6. Pattern Piece Information Block

Every pattern piece must include:

1. **Piece identifier**: Number and name (e.g., "1 – Front Bodice")
2. **Size**: Measurement set identifier
3. **Cutting instructions**: Quantity and fabric type
4. **Date**: Drafting date (ISO format: YYYY-MM-DD)
5. **Scale verification**: 5cm x 5cm test square

1.7. Color Standards

1.7.1. Fabric Type Colors

Color Name	Hex Code	R Variable	Usage
Primary fabric	#212121	`fabric_primary`	Main garment fabric
Lining	#1976D2	`fabric_lining`	Lining pieces
Interfacing	#F57C00	`fabric_interfacing`	Interfacing/fusing

1.7.2. Annotation Colors

Purpose	Hex Code	R Variable
Seam allowance	#757575	`seam_allowance`
Grainline	#388E3C	`grainline`
Cut on fold	#D32F2F	`cut_on_fold`
Dimensions	#1565C0	`dimensions`
Front notch	#212121	`notch_default`
Back notch	#D32F2F	`notch_back`

1.8. Grid System

All pattern plots include a reference grid for visual measurement:

- **Major grid lines**: Every 5 cm, solid light grey (#E0E0E0), linewidth 0.3
- **Minor grid lines**: Every 1 cm, very light grey (#F5F5F5), linewidth 0.1
- **Axis labels**: Centered on each major grid line

The grid serves as: - Quick visual reference for dimensions - Scale verification tool - Alignment guide for pattern pieces

1.9. File Naming Conventions

```
{garment}_{block}_{version}_{size}.{ext}

Examples:
  bodice_front_female_v1.0_size48.pdf
  sleeve_basic_v2.1_universal.csv
  skirt_pencil_variation_v1.0_size42_A0.pdf
```

1.9.1. Version Numbering

- **Major** (X.0.0): Complete redraft or design change
- **Minor** (0.X.0): Significant adjustments to fit
- **Patch** (0.0.X): Small corrections, annotation fixes

1.10. Printing Standards

1.10.1. Scale Verification

Always print and measure the 5cm test square before cutting. If the square doesn't measure exactly 5cm x 5cm: 1. Check printer settings for "Actual size" (not "Fit to page") 2. Verify PDF scaling is set to 100% 3. Re-print if scale is off by more than 1mm

1.10.2. Paper Sizes

Format	Dimensions (cm)	Use Case
A4	21.0 x 29.7	Home printing (tiled)
A3	29.7 x 42.0	Small patterns
A0	84.1 x 118.9	Plotter, full patterns

1.10.3. Tile Assembly

For tiled home printing: 1. Trim along registration marks 2. Align matching marks between tiles 3. Tape from center outward 4. Verify scale after assembly

1.11. Measurement File Format

1.11.1. SeamlyMe CSV Structure

```
code,reference,description,value,formula
height,A01,Height: Total,154,154
bust_circ,G04,Bust circumference,113,113
```

1.11.2. Ease Parameters CSV Structure

```
parameter,value,unit,description
bust_ease,6,cm,Overall bust ease
shoulder_slope_front,4,cm,Front shoulder drop
```

1.12. R Implementation Notes

1.12.1. Function Naming

- `calc_*`: Pure mathematical calculations (distance, angle)
- `create_*`: Generate new objects (curves, plots)

- `add_*`: Modify existing plots with annotations
- `read_*` / `write_*`: File I/O operations
- `validate_*`: Check data integrity

1.12.2. Coordinate Storage

- Points stored as named vectors: `c(x, y)`
- Point collections as named lists: `list(A = c(0, 44), B = c(48, 44))`
- Plotted points as data frames: `data.frame(name, x, y)`

1.12.3. Cache Strategy

- All plots cached in `images/cache/`
- Cache invalidation based on parameter hash
- Manual cache clear: delete `images/cache/*.pdf`

1.13. References

- FreeSewing Notation Guide: https://freesewing.eu/docs/about/notation/
- ISO 8559: Garment construction and anthropometric surveys
- ABNT NBR 13377: Brazilian standard for garment sizes (where applicable)

2. Measurement Systems and Integration

This chapter covers how body measurements are captured, stored, and integrated into the parametric pattern drafting system. We use SeamlyMe as the primary measurement tool, but the system accepts any CSV following our format.

2.1. Measurement Philosophy

2.1.1. Body Measurements vs Pattern Measurements

Body measurements are taken directly from the body (or a dress form). They represent the actual dimensions of the person who will wear the garment.

Pattern measurements are body measurements plus ease. They represent the dimensions of the finished garment.

Pattern Measurement = Body Measurement + Ease

2.1.2. Types of Ease

Ease Type	Purpose	Example
Wearing ease	Minimum needed for movement and comfort	4-6 cm at bust
Design ease	Style choice affecting silhouette	Additional 10 cm for oversized look
Negative ease	Garment smaller than body (stretch fabrics)	-2 cm at bust for fitted knit

2.2. SeamlyMe Integration

SeamlyMe is open-source software for managing body measurements. It exports measurements in a CSV format we can read directly.

2.2.1. Export Process from SeamlyMe

1. Open your measurement file (`.vit`) in SeamlyMe
2. Go to **Measurements → Export to CSV**
3. Select all measurements needed for the pattern
4. Save the CSV to `data/measurements/`

2.2.2. SeamlyMe CSV Format

```
code,reference,description,value,formula
height,A01,Height: Total,154,154
bust_circ,G04,Bust circumference,113,113
neck_front_to_waist_f,H01,Neck Front to Waist Front,58,58
```

Columns explained:

```
code: Internal identifier used in our R code (e.g., bust_circ)

reference: SeamlyMe reference code (e.g., G04)

description: Human-readable description
```

`value`: The actual measurement in centimeters

`formula`: Can be a fixed value or a formula referencing other measurements

Required Measurements by Block Type

Female Bodice Block Requirements

3. Required measurements for female bodice

```r
female_bodice_required <- c(
  "height",
  "bust_circ",
  "neck_front_to_waist_f",
  "back_waist_length",
  "shoulder_length",
  "armscye_circ",
  "neck_width",
  "across_chest_f",
  "bustpoint_to_bustpoint",
  "bustpoint_to_neck_side"
)
```

4. Create reference table

```r
reqs_df <- data.frame(
  Code = female_bodice_required,
  Description = c(
    "Total height",
    "Bust circumference",
    "Front waist length (neck to waist)",
    "Back waist length",
    "Shoulder length",
    "Armscye circumference",
    "Neck width",
    "Across chest (front)",
    "Bust point to bust point",
    "Bust point to neck side"
  ),
  Category = c("Reference", "Circumference", "Length", "Length",
               "Length", "Circumference", "Width", "Width",
               "Width", "Length")
)

knitr::kable(reqs_df,
             caption = "Required measurements for female bodice block",
             booktabs = TRUE) %>%
  kable_styling(latex_options = c("hold_position", "striped"))
```

Male Bodice Block Requirements

```r
male_bodice_required <- c(
  "height",
  "chest_circ",
  "neck_front_to_waist_f",
  "back_waist_length",
  "shoulder_length",
  "armscye_circ",
  "neck_width",
  "across_back",
  "across_chest_m"
)

reqs_male_df <- data.frame(
  Code = male_bodice_required,
  Description = c(
    "Total height",
    "Chest circumference",
    "Front waist length",
    "Back waist length",
    "Shoulder length",
    "Armscye circumference",
    "Neck width",
```

```r
    "Across back width",
    "Across chest (male)"
  ),
  Category = c("Reference", "Circumference", "Length", "Length",
               "Length", "Circumference", "Width", "Width", "Width")
)

knitr::kable(reqs_male_df,
             caption = "Required measurements for male bodice block",
             booktabs = TRUE) %>%
  kable_styling(latex_options = c("hold_position", "striped"))
```

Skirt Block Requirements

```r
skirt_required <- c(
  "waist_circ",
  "hip_circ",
  "waist_to_hip",
  "waist_to_knee",
  "height"
)

reqs_skirt_df <- data.frame(
  Code = skirt_required,
  Description = c(
    "Waist circumference",
    "Hip circumference",
    "Waist to hip distance",
    "Waist to knee distance",
    "Total height"
  ),
  Category = c("Circumference", "Circumference", "Length", "Length", "Reference")
)

knitr::kable(reqs_skirt_df,
             caption = "Required measurements for skirt block",
             booktabs = TRUE) %>%
  kable_styling(latex_options = c("hold_position", "striped"))
```

Table 4.1: Required measurements for skirt block

Code	Description	Category
waist_circ	Waist circumference	Circumference
hip_circ	Hip circumference	Circumference
waist_to_hip	Waist to hip distance	Length
waist_to_knee	Waist to knee distance	Length
height	Total height	Reference

Sleeve Block Requirements

```r
sleeve_required <- c(
  "armscye_circ",
  "arm_upper_circ",
  "arm_elbow_circ",
  "arm_wrist_circ",
  "arm_shoulder_tip_to_wrist",
  "arm_shoulder_tip_to_elbow",
  "arm_armpit_to_wrist"
)

reqs_sleeve_df <- data.frame(
  Code = sleeve_required,
  Description = c(
    "Armscye circumference",
    "Upper arm circumference",
    "Elbow circumference",
    "Wrist circumference",
    "Shoulder to wrist length",
    "Shoulder to elbow length",
    "Armpit to wrist (inside)"
  ),
  Category = c("Circumference", "Circumference", "Circumference",
               "Circumference", "Length", "Length", "Length")
)

knitr::kable(reqs_sleeve_df,
             caption = "Required measurements for sleeve block",
             booktabs = TRUE) %>%
  kable_styling(latex_options = c("hold_position", "striped"))
```

Table 4.2: Required measurements for sleeve block

Code	Description	Category
armscye_circ	Armscye circumference	Circumference
arm_upper_circ	Upper arm circumference	Circumference
arm_elbow_circ	Elbow circumference	Circumference
arm_wrist_circ	Wrist circumference	Circumference
arm_shoulder_tip_to_wrist	Shoulder to wrist length	Length
arm_shoulder_tip_to_elbow	Shoulder to elbow length	Length
arm_armpit_to_wrist	Armpit to wrist (inside)	Length

Pants Block Requirements

```r
pants_required <- c(
  "waist_circ",
  "hip_circ",
  "waist_to_hip",
  "crotch_depth",
  "inseam",
  "outseam",
  "thigh_circ",
  "knee_circ",
```

```
  "ankle_circ"
)

reqs_pants_df <- data.frame(
  Code = pants_required,
  Description = c(
    "Waist circumference",
    "Hip circumference",
    "Waist to hip distance",
    "Crotch depth (sitting)",
    "Inseam length",
    "Outseam length",
    "Thigh circumference",
    "Knee circumference",
    "Ankle circumference"
  ),
  Category = c("Circumference", "Circumference", "Length",
               "Length", "Length", "Length", "Circumference",
               "Circumference", "Circumference")
)

knitr::kable(reqs_pants_df,
             caption = "Required measurements for pants block",
             booktabs = TRUE) %>%
  kable_styling(latex_options = c("hold_position", "striped"))
```

Table 4.3: Required measurements for pants block

Code	Description	Category
waist_circ	Waist circumference	Circumference
hip_circ	Hip circumference	Circumference
waist_to_hip	Waist to hip distance	Length
crotch_depth	Crotch depth (sitting)	Length
inseam	Inseam length	Length
outseam	Outseam length	Length
thigh_circ	Thigh circumference	Circumference
knee_circ	Knee circumference	Circumference
ankle_circ	Ankle circumference	Circumference

Loading Measurements in R

The `read_measurements()` Function

```
# Source measurement functions
# source(here::here("Rscripts", "measurement_io.R"))

# Load a measurement file
fem48 <- read_measurements(here::here("data", "measurements", "female_size48.csv"))

# Display key measurements
key_codes <- c("height", "bust_circ", "neck_front_to_waist_f", "shoulder_length")

cat("Key measurements for female size 48:\n")
```

```
## Key measurements for female size 48:
```

```r
for (code in key_codes) {
  cat(sprintf("  %-30s: %6.1f cm\n", code, m_get(fem48, code)))
}
```

```
##   height                         :  154.0 cm
##   bust_circ                      :  113.0 cm
##   neck_front_to_waist_f          :   58.0 cm
##   shoulder_length                :   10.5 cm
```

Creating a Measurement Template

If you don't have SeamlyMe, you can create a template CSV and fill in values manually:

```r
# Create template for manual editing
create_measurement_template(
  filepath = "data/measurements/custom_measurements.csv",
  codes = c("height", "bust_circ", "waist_circ", "hip_circ",
            "neck_front_to_waist_f", "back_waist_length",
            "shoulder_length", "armscye_circ", "neck_width")
)
```

Measurement Validation

Checking Required Measurements

```r
# Validate that all required measurements are present
tryCatch({
  validate_measurements(fem48, female_bodice_required)
  cat("  All required measurements present\n")
}, error = function(e) {
  cat(" ", e$message, "\n")
})
```

```
##   All required measurements present
```

```r
# Check for zero values (likely unfilled)
zero_measurements <- names(fem48)[fem48 == 0]
if (length(zero_measurements) > 0) {
  cat("\n  Zero values found in:", paste(zero_measurements, collapse = ", "))
  cat("\n  These should be measured and filled in before drafting.\n")
}
```

```r
# Logical consistency checks
checks <- list()

# Waist should be smaller than bust
if (!is.null(fem48$waist_circ) && !is.null(fem48$bust_circ)) {
  checks$waist_vs_bust <- fem48$waist_circ < fem48$bust_circ
}

# Hip should be larger than waist
```

```r
if (!is.null(fem48$hip_circ) && !is.null(fem48$waist_circ)) {
  checks$hip_vs_waist <- fem48$hip_circ > fem48$waist_circ
}

# Arm length checks
if (!is.null(fem48$arm_shoulder_tip_to_wrist) &&
    !is.null(fem48$arm_shoulder_tip_to_elbow)) {
  checks$arm_segments <- fem48$arm_shoulder_tip_to_elbow <
                          fem48$arm_shoulder_tip_to_wrist
}

cat("Consistency checks:\n")
```

```
## Consistency checks:
```

```r
for (check_name in names(checks)) {
  status <- if (checks[[check_name]]) " " else " "
  cat(sprintf("  %s %s\n", status, check_name))
}
```

```
##     waist_vs_bust
##     hip_vs_waist
##     arm_segments
```

Ease Parameters

Ease parameters are stored separately from body measurements.

This separation allows:

```
Changing body measurements without affecting design decisions

Experimenting with different fits on the same body

Sharing ease settings across measurement sets
```

Ease Parameters File Format

```r
# Read ease parameters
bodice_ease <- read_ease(here::here("data", "parameters", "bodice_ease_female.csv"))

# Display as table
ease_df <- data.frame(
  Parameter = names(bodice_ease),
  Value = paste(unlist(bodice_ease), "cm"),
  Description = c(
    "Overall bust circumference ease",
    "Waist circumference ease",
    "Hip circumference ease",
    "Front shoulder drop from J point",
    "Back neckline drop from B",
    "Reference armscye depth",
    "Front neckline lateral shift",
```

```
    "Back neckline lateral shift",
    "Back shoulder dart height",
    "Back shoulder extension",
    "Side seam horizontal offset"
  )
)

knitr::kable(ease_df,
             caption = "Female bodice ease parameters",
             booktabs = TRUE) %>%
  kable_styling(latex_options = c("hold_position", "striped"))
```

Ease by Garment Type and Fit Garment Close Fit Semi-Fit Loose Fit Oversized Bodice bust 4-6 cm 6-8 cm 8-12 cm 12-20 cm Bodice waist 2-4 cm 4-6 cm 6-10 cm 10-16 cm Skirt hip 2-4 cm 4-6 cm 6-8 cm 8-12 cm Pants hip 2-4 cm 4-6 cm 6-8 cm — Sleeve biceps 2-4 cm 4-6 cm 6-8 cm 8-12 cm Standard Size Reference Tables Female Standard Measurements (ABNT NBR 13377)

```
female_standard <- data.frame(
  Size = c("38", "40", "42", "44", "46", "48", "50"),
  Bust = c(84, 88, 92, 96, 102, 108, 114),
  Waist = c(64, 68, 72, 76, 82, 88, 94),
  Hip = c(92, 96, 100, 104, 110, 116, 122),
  Neck_to_Waist_F = c(44, 45, 46, 47, 48, 50, 52),
  Back_Waist = c(38, 39, 40, 41, 42, 43, 44)
)

knitr::kable(female_standard,
             caption = "Standard female measurements (cm) - Reference only",
             booktabs = TRUE) %>%
  kable_styling(latex_options = c("hold_position", "striped"))
```

Table 4.4: Standard female measurements (cm) — Reference only

Size	Bust	Waist	Hip	Neck_to_Waist_F	Back_Waist
38	84	64	92	44	38
40	88	68	96	45	39
42	92	72	100	46	40
44	96	76	104	47	41
46	102	82	110	48	42
48	108	88	116	50	43
50	114	94	122	52	44

Male Standard Measurements

```
male_standard <- data.frame(
  Size = c("46", "48", "50", "52", "54", "56"),
  Chest = c(92, 96, 100, 104, 108, 112),
  Waist = c(78, 82, 86, 90, 96, 102),
  Hip = c(94, 98, 102, 106, 110, 114),
  Back_Waist = c(44, 45, 46, 47, 48, 49)
)

knitr::kable(male_standard,
```

```
             caption = "Standard male measurements (cm) - Reference only",
             booktabs = TRUE) %>%
  kable_styling(latex_options = c("hold_position", "striped"))
```

Table 4.5: Standard male measurements (cm) — Reference only

Size	Chest	Waist	Hip	Back_Waist
46	92	78	94	44
48	96	82	98	45
50	100	86	102	46
52	104	90	106	47
54	108	96	110	48
56	112	102	114	49

Practical Measurement Tips Taking Body Measurements

```
Use a flexible tape measure - not a metal one

Measure over undergarments - or the clothing layer the garment will be worn over

Keep the tape parallel to the floor - especially for circumferences

Don't pull too tight - the tape should be snug but not compressing

Mark landmarks - use elastic bands to mark waist, hip, and bust levels

Take multiple measurements - average three readings for accuracy

Record posture - note if the subject has forward shoulders, sway back, etc.
```

Measurement Landmarks Landmark How to Find Waist level Side bend — where the body creases Hip level Widest part of hips, usually 18-22 cm below waist Bust point Apex of bust (nipple level) Neck base Where neck meets shoulder Shoulder tip Acromion bone (bony point at shoulder edge) Elbow Olecranon process (bony point of elbow) Wrist Styloid process (bony bump at wrist) Chapter Summary

```
All measurements in centimeters, stored in SeamlyMe CSV format

Measurements are validated for presence and consistency before drafting

Ease parameters are stored separately for flexibility

Standard size tables provided for reference only - use actual body measurements when possible

Each block type has specific measurement requirements documented above
```

5. Curso por Correspondência — Sophia Jobim

Referência: JOBIM, Sophia. *O sistema de corte e costura de Sophia Jobim: os anos de ouro de Mme Carvalho no Liceu Império: 1932-1954*. São Paulo: Portal de Livros Abertos da USP, 2024. Disponível em: https://www.livrosabertos.abcd.usp.br/portaldelivrosUSP/catalog/view/1327/1210/4651

Este capítulo documenta o estudo do método de modelagem de Sophia Jobim, transcrevendo as instruções originais e implementando-as em código R parametrizado.

5.1. Blusa (Corpo Simples)

Página de referência: ~111 do PDF original

5.1.1. Dados de Entrada

Para um primeiro teste vamos usar as medidas padrão do livro para o manequim 48.

5.1.1.1. Medidas do Manequim 48

```r
# Carregar medidas do arquivo CSV
medidas_sj <- read_measurements(here("data", "measurements", "sophia_jobim_size48.csv"))

# Extrair medidas relevantes
medida_busto <- m_get(medidas_sj, "bust_circ")
comp_frente <- m_get(medidas_sj, "neck_front_to_waist_f")
comp_costas <- m_get(medidas_sj, "back_waist_length")
grossura_braco <- m_get(medidas_sj, "armscye_circ")

# Exibir tabela de medidas
medidas_df <- data.frame(
  Codigo = c("bust_circ", "neck_front_to_waist_f", "back_waist_length", "armscye_circ"),
  Descricao = c("Circunferencia do busto", "Comprimento frente", "Comprimento costas", "Grossura do braco"),
  Valor_cm = c(medida_busto, comp_frente, comp_costas, grossura_braco)
)

knitr::kable(medidas_df,
             caption = "Medidas do manequim 48 (Sophia Jobim)",
             booktabs = TRUE) %>%
  kable_styling(latex_options = c("hold_position", "striped"))
```

Table 5.1: Medidas do manequim 48 (Sophia Jobim)

Codigo	Descricao	Valor_cm
bust_circ	Circunferencia do busto	90
neck_front_to_waist_f	Comprimento frente	44
back_waist_length	Comprimento costas	41
armscye_circ	Grossura do braco	33

5.1.1.2. Parametros de Folga e Construcao

```
# Carregar parametros de ease
ease_sj <- read_ease(here("data", "parameters", "sophia_jobim_ease.csv"))

# Exibir tabela de parametros
ease_df <- data.frame(
  Parametro = names(ease_sj),
  Valor = paste(unlist(ease_sj), "cm")
)

knitr::kable(ease_df,
             caption = "Parametros de folga e construcao (Sophia Jobim)",
             booktabs = TRUE) %>%
  kable_styling(latex_options = c("hold_position", "striped"))
```

Table 5.2: Parametros de folga e construcao (Sophia Jobim)

Parametro	Valor
bust_ease	6 cm
side_seam_offset	3 cm
shoulder_slope_front	4 cm
front_neckline_depth_extra	2 cm
front_neckline_shift	1 cm
back_neck_drop	2 cm
back_neckline_shift	1 cm
back_dart_height	5 cm
back_shoulder_extension	1 cm
bissetriz_R	1.5 cm
bissetriz_S	2.5 cm
desvio_V_esquerda	2 cm
desvio_X_esquerda	0.5 cm
subida_inferior	3 cm
desvio_W_lateral	2 cm

5.1.1.3. Tabela de Profundidade da Cava

```
# Carregar tabela de profundidade da cava
tabela_cava <- read_csv(here("data", "measurements", "sophia_jobim_armscye_table.csv"),
                        show_col_types = FALSE)

# Encontrar profundidade para a grossura do braco
profundidade_cava <- tabela_cava$armscye_depth[tabela_cava$arm_circ == grossura_braco]

if (length(profundidade_cava) == 0) {
  profundidade_cava <- approx(tabela_cava$arm_circ, tabela_cava$armscye_depth,
                              xout = grossura_braco)$y
}

# Destacar a linha usada
tabela_cava$Destaque <- ifelse(tabela_cava$arm_circ == grossura_braco, "<- USADO", "")

knitr::kable(tabela_cava,
             caption = "Tabela 1: Profundidade da cava conforme grossura do braco",
```

```
              booktabs = TRUE) %>%
  kable_styling(latex_options = c("hold_position", "striped"))
```

Table 5.3: Tabela 1: Profundidade da cava conforme grossura do braco

arm_circ	armscye_depth	Destaque
29	18.0	
30	19.0	
31	19.5	
32	20.0	
33	21.0	<- USADO
34	22.0	
35	22.5	
36	23.0	
37	23.5	
38	24.0	
39	24.5	
40	25.0	
41	25.5	
42	26.0	
43	26.5	
44	27.0	
45	27.5	
46	28.0	

```
cat(sprintf("\n**Braco:** %.0f cm -> **Profundidade da cava:** %.0f cm\n",
            grossura_braco, profundidade_cava))
```

```
##
## **Braco:** 33 cm -> **Profundidade da cava:** 21 cm
```

5.1.2. Instrucoes Originais (transcricao)

> Para facilitar as explicacoes, tomarei como exemplo as medidas do manequim 48, que sao as mais aproximadas do corpo medio feminino. Essas medidas sao, para a blusa: 90 cm para a circunferencia do busto; 44 cm para o comprimento da blusa, na frente, e 41 cm para o comprimento nas costas.
>
> Para que a blusa nao fique justa como um colete, torna-se necessario dar-lhe um aumento de 6 cm na circunferencia do busto, resultando 96. Como vamos fazer apenas a metade do molde, isto e, metade da frente e metade das costas, dividiremos por 2 a medida total 96 cm. Que resulta 48 cm.

5.1.3. Execucao do Tracado

```
# Executar o tracado com a nova funcao modular
blusa <- draft_blouse_sj(
  medidas = medidas_sj,
  ease = ease_sj,
  armscye_depth = profundidade_cava
)
```

```r
cat(sprintf("
**Dimensoes do molde:**
- Largura total: %.1f cm
- Altura total: %.1f cm
- Divisao da frente: %.1f cm
- Divisao das costas: %.1f cm
", blusa$dimensions$dim_width,
   blusa$dimensions$dim_height,
   blusa$dimensions$dim_front_division,
   blusa$dimensions$dim_back_division))
```

```
##
## **Dimensoes do molde:**
## - Largura total: 48.0 cm
## - Altura total: 44.0 cm
## - Divisao da frente: 6.4 cm
## - Divisao das costas: 5.6 cm
```

5.1.4. Molde Completo

```r
# Usar a funcao de plot do modulo
plot_pattern_notebook(
  block = blusa,
  filename = "blusa_sj_final",
  title = "Blusa (Corpo Simples) - Metodo Sophia Jobim",
  subtitle = paste("Manequim 48 | Busto:", medida_busto, "cm"),
  show_points = TRUE,
  show_grid = TRUE,
  show_construction = TRUE,
  piece_type = "both",
  width = 10,
  height = 10
)
```

5.1.5. Comprimento das Linhas

```r
comprimentos_df <- data.frame(
  Segmento = c(
    "Ombro frente (I-M)",
    "Decote frente (curva N-I2)",
    "Ombro costas (L-P)",
    "Decote costas (curva B'-L2)",
    "Cava (curva P-M)",
    "Lateral frente (Q-c)",
    "Lateral costas (Q-d)",
    "Cintura frente (c-b)",
```

```r
    "Cintura costas (d-a)",
    "Barra frente (b-D)",
    "Centro frente (D-N)",
    "Centro costas (a-B')"
  ),
  Comprimento_cm = round(c(
    blusa$line_lengths$shoulder_front_IM,
    blusa$line_lengths$neckline_front,
    blusa$line_lengths$shoulder_back_LP,
    blusa$line_lengths$neckline_back,
    blusa$line_lengths$armscye,
    blusa$line_lengths$side_front_Qc,
    blusa$line_lengths$side_back_Qd,
    blusa$line_lengths$waist_front_cb,
    blusa$line_lengths$waist_back_da,
    blusa$line_lengths$hem_front_bD,
    blusa$line_lengths$center_front_DN,
    blusa$line_lengths$center_back_aB
  ), 1)
)

knitr::kable(comprimentos_df,
             caption = "Comprimento das linhas que delimitam o molde",
             booktabs = TRUE) %>%
  kable_styling(latex_options = c("hold_position", "striped"))
```

Table 5.4: Comprimento das linhas que delimitam o molde

Segmento	Comprimento_cm
Ombro frente (I-M)	13.4
Decote frente (curva N-I2)	11.4
Ombro costas (L-P)	13.3
Decote costas (curva B'-L2)	7.9
Cava (curva P-M)	41.1
Lateral frente (Q-c)	20.1
Lateral costas (Q-d)	20.1
Cintura frente (c-b)	17.4
Cintura costas (d-a)	20.5
Barra frente (b-D)	6.4
Centro frente (D-N)	35.6
Centro costas (a-B')	39.0

5.1.6. Area Aproximada para Calculo de Tecido

```r
area_info <- blusa$area

area_df <- data.frame(
  Peca = c("Frente", "Costas", "Total (2 pecas)"),
  Area_cm2 = c(area_info$front_area_cm2, area_info$back_area_cm2, area_info$total_area_cm2),
  Area_m2 = c(area_info$front_area_m2, area_info$back_area_m2, area_info$total_area_m2)
```

```r
)

knitr::kable(area_df,
             caption = "Area aproximada das pecas do molde",
             booktabs = TRUE) %>%
  kable_styling(latex_options = c("hold_position", "striped"))
```

Table 5.5: Area aproximada das pecas do molde

Peca	Area_cm2	Area_m2
Frente	871.2419	0.0871242
Costas	767.7236	0.0767724
Total (2 pecas)	1638.9655	0.1638965

```r
cat(sprintf("
**Estimativa de tecido:**
- Area total do molde (frente + costas): %.2f m2
- Com margem de 15%% para curvas e costuras: %.2f m2
- Tecido recomendado (largura 1.40 m): %.2f m linear

> **Nota:** %s
", area_info$total_area_m2,
   area_info$total_area_m2 * 1.15,
   (area_info$total_area_m2 * 1.15) / 1.40,
   area_info$note))
```

```
##
## **Estimativa de tecido:**
## - Area total do molde (frente + costas): 0.16 m2
## - Com margem de 15% para curvas e costuras: 0.19 m2
## - Tecido recomendado (largura 1.40 m): 0.13 m linear
##
## > **Nota:** Area aproximada. Adicionar 10-15% para margens e curvas.
```

5.1.7. Nota Tecnica: Escolha do Metodo para a Curva da Cava

1. Spline com pontos de controle manuais

Tentativa de ajustar a curva usando pontos de controle arbitrarios. Resultado: zigue-zague e barriga invertida no decote. Descartado.

2. Bezier cubica por segmentos

Dois segmentos (frente: M-Y-T-Q, costas: Q-U-Z-P) com pontos de controle calculados para t=1/3 e t=2/3. Resultado: passou sobre os pontos mas apresentou transicao brusca em Q. Descartado.

3. LOESS (Local Polynomial Regression)

Regressao polinomial local com span entre 0.3 e 0.7. Resultado: curvas muito suaves e organicas, mas nao passam exatamente sobre os pontos intermediarios (Z, U, T, Y). Descartado.

4. Spline cubica parametrica (ESCOLHIDO)

Spline cubica natural usando distancia acumulada como parametro. Passa exatamente sobre todos os 7 pontos (P, Z, U, Q, T, Y, M). Minimiza a curvatura total, garantindo maxima suavidade matematica. A condicao natural nos extremos permite curvatura zero em P e M. Resolve o problema de ordenacao espacial usando o parametro t baseado em distancia acumulada.

Decisao: O metodo escolhido foi a spline cubica parametrica por atender simultaneamente aos dois requisitos: passagem exata sobre todos os pontos de construcao e maxima suavidade da curva.

5.1.8. Exportar para Tamanho Real

```
plot_pattern_fullscale(
  block = blusa,
  output_file = "output/blusa_corpo_simples_A0.pdf",
  paper_size = "A0",
  include_seam_allowance = TRUE,
  seam_allowance = 1.5
)
```

5.1.9. Resumo do Aprendizado

1. **Metodo Sophia Jobim**: usa divisoes proporcionais baseadas no busto
2. **Divisoes**: I e J na frente (1/4 e 3/4 de AG), K e L nas costas (1/4 e 3/4 de GB)
3. **Cava**: construida com spline cubica parametrica sobre 7 pontos (P, Z, U, Q, T, Y, M)
4. **Decote frente**: profundidade = AI + 2 cm, com desvio de 1 cm sobre linha do ombro (ponto I2)
5. **Decote costas**: desce 2 cm de B, com desvio de 1 cm sobre linha do ombro (ponto L2)
6. **Costas**: sempre 3 cm mais estreitas que a frente (deslocamento EF)
7. **Ombro costas**: tem pence de 5 cm em K, prolongada 1 cm para P
8. **Peca das costas**: fecha com pontos a (barra), subindo ate B' (decote costas)
9. **Area aproximada**: calculada por poligono sem curvas; adicionar 15% para margem de costura